

Bitcoin, Blockchain and Cryptoassets

Problem Set

April 22, 2026

In this problem set, you will convert between different numeral systems, attempt to break a symmetric cipher, use the RSA algorithm to encrypt and decrypt a message, and analyse transaction data with a block explorer.

This problem set will be graded. By completing the following problems, you can earn up to **4 bonus points** for the exam. For each problem, you will receive **1 point** for a completely correct solution and **0.5 points** for a partially correct solution.

Please submit your solutions by **May 24th, 2026 (end of day)** via email to patrik.haenggi@unibas.ch. Be sure to use your **university email address** and include **"Submission Bonus Points"** in the subject line.

At the beginning of your email, clearly state your **full name** and **student ID (Matrikelnummer)**.

Submissions received after the deadline will not be considered. If you have any questions regarding the content, please contact patrik.haenggi@unibas.ch.

Problem 1

This first problem aims to familiarize you with the conversion between different numeral systems.

Problem 1.1

- a) Convert the binary number $(10010110)_{B=2}$ into a decimal number.
- b) Convert the decimal number $(77)_{B=10}$ into a binary number.

Problem 1.2

- a) Convert the binary number $(11011001)_{B=2}$ into a hexadecimal (with basis $B = 16$).
- b) Convert the hexadecimal $(4d7a)_{B=16}$ into a binary number.

Problem 2

The following (English) text was encrypted with the monoalphabetic substitution cipher:

LNCPUVBLUNF, YKNZCZSBUF, BFQ XHBWL ZNFLWBZLX BWP LPWHX
LSBL POPWTNFP UX QUXZIXXUFM. LSP ZPFLPW GNW UFFNOBLUOP
GUFBFZP BL LSP IFUOPWXULT NG YBXPK NGGPWX LSP IFUAIP ZS-
BFZP LN ZNFOPWL LSPNWPLUZBK CFNEKPQMP BFQ UFFNOBLUOP
UQPBX UFLN JWBZLUZBK BJJKUZBLUNFX UF LSP BWPB NG YKNZCZS-
BUFX EULS LSP YKNZCZSBUF ZSBKKPFMP. BZZNHJBFUPQ YT PRJP-
WLX BFQ ZNBZSPX, LSP XLIQPFLX MN LSWNIMS LSP HNXL UHJN-
WLBFL XLPJX UF ZWPBLUFM B XNKILUNF ZNFZPJL, UFZKIUUFM LPZS-
FUZBK JWNLNLTJPX. LSP ZNIWXP UX NGGPWPQ BX B LEN EPPC YKNZC
XPHUFBW, LBCUFM JKBZP YPGNWP LSP XLBWL NG LSP GBKK XPH-
PXLPW.

Decrypt it using frequency analysis and by exploiting the relations between letters. (Hint: Use frequency analysis only to decipher the three most frequent letters. Also, there are online tools that can help you find the relative frequencies of the letters¹).

¹E.g. <https://www.101computing.net/frequency-analysis/>

Problem 3

In this problem, we use the RSA algorithm covered in the lecture on asymmetric cryptography, to create a private and public key and to send an encrypted message similar to the example in the lecture. We use the parameters $p = 13$, $q = 29$ and $e = 5$ to encipher the message L .

Problem 3.1

Compute the number N and convert the message $M = L$ using the ASCII table into a decimal number (!). Subsequently, encrypt this decimal number to receive the encrypted message C .²

Problem 3.2

Compute the number $\phi(N)$ and derive the private key k_p , which will be used to decrypt the message C .

Hint: To find the multiplicative inverse in modulo-calculations we have to use the Euclidean Algorithm. There are several online tools that you can use instead of doing this by hand³.

Problem 3.3

Decrypt the encrypted message C using the private key k_p and check the result with the original message $M = L$.

²Conventional calculators might not be able to handle numbers of this size well. We recommend using the online modulo calculator of <https://planetcalc.com/8326/>.

³A simple tool would be the inverse calculator by <https://planetcalc.com/3311/>. To derive the private key (k_p) in the RSA example in the lecture on asymmetric cryptography, you would have to insert 7 under "Integer" and 160 under "Modulo". The result is the private key k_p . Use the same procedure in this problem.

Problem 4

Problem 4 revolves around Bitcoin transaction data. Use a block explorer⁴ to answer the following questions.

- a) Find the transaction that was initiated by the address

`1CGN1zd62UakPw7neohJN8zTJUJ9LT2Ay8`

in block 325,001. What is the transaction hash?

- b) Who mined block 325,001 and what is the Bitcoin address?
- c) How many in- and outputs does the transaction in a) consist of?
- d) What is the `pubKeyHash` of the output with index 4?
- e) One output has a different script transaction type than the others. What is it and how is it used?

⁴E.g. <https://www.blockchain.com/explorer>